Number Theoretic Transform (NTT)

October 29, 2026

Blockstream Research

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github.com/BlockstreamResearch/zk-book



Plan

1. Recap on Interpolation
   - Polynomial Interpolation is a Universal Encoder
   - Motivation for NTT
2. Roots of Unity
   - Multiplicative Subgroup of Finite Fields
   - Barycentric Interpolation
3. Number Theoretic Transform
   - Three Gadgets
   - Polynomial vs NTT Domain
4. Details
   - Why NTT takes quasilinear complexity?

Recap on Interpolation

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All the previous protocols use the idea that polynomials are **universal data encoders**. We can encode any set of scalars $(a_0, \ldots, a_{N-1}) \in \mathbb{F}^N$ using **interpolation**:

$$p(x_j) = a_j, \quad j \in [N], \quad \{x_j\}_{j\in[N]} \text{ are fixed}$$

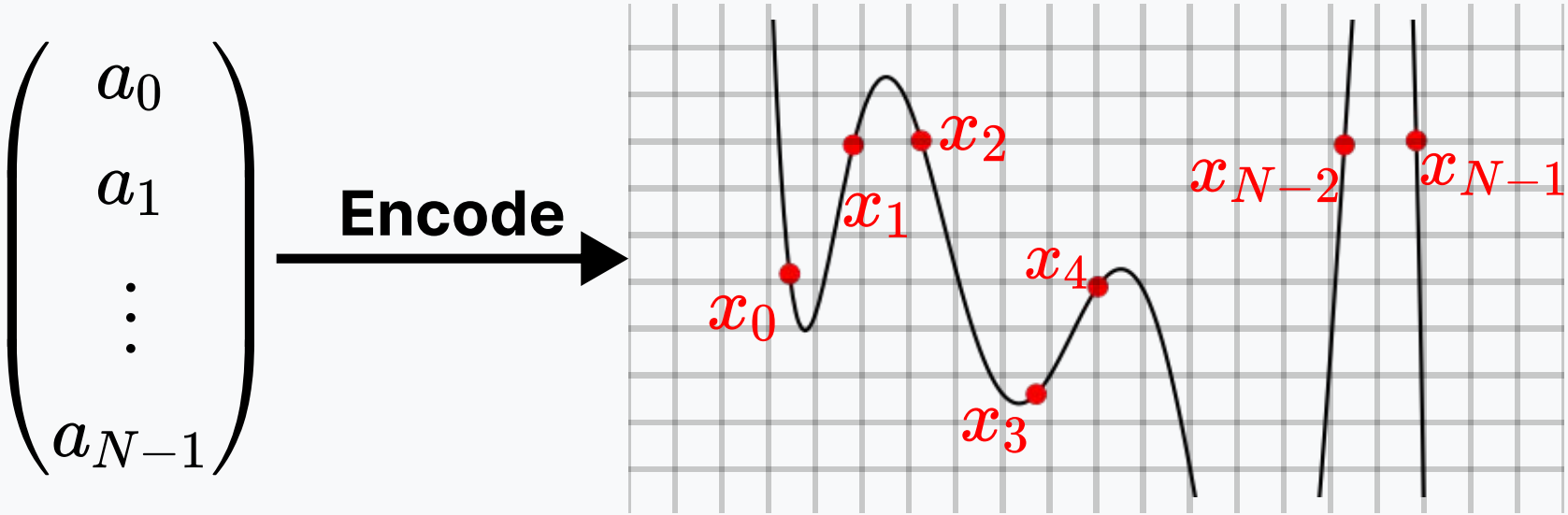


Figure: Polynomial Interpolation as a universal encoder.

Polynomial Interpolation

Example

In Groth16, we used interpolation of $3n$ polynomials:

$$\ell_j(i) = L_{i,j}, \quad r_j(i) = R_{i,j}, \quad o_j(i) = O_{i,j},$$

where $L_{i,j}, R_{i,j}, O_{i,j}$ are the elements of constraint matrices L, R, O (left, right, and output).

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However, in PlonK (see the book) we will interpolate $a(\omega^j) = A_j$ where A_j are the elements of the left trace vector A.

Question

What the heck is this ω? Why do we need it? How does it help?

A	B	C
2	3	6
6	3	9
9	✗	8

$a(1) = 2$ $b(1) = 3$ $c(1) = 6$
$a(\omega) = 6$ $b(\omega) = 3$ $c(\omega) = 9$
$a(\omega^2) = 9$ $b(\omega^2) = 0$ $c(\omega^2) = 8$

$a(x)$ $b(x)$ $c(x)$

Why we need something advanced?

Recall

The interpolation formula is given by:

$$p(T) = \sum_{i=0}^{N-1} a_i \cdot \ell_i(T), \quad \ell_i(T) = \prod_{j=0, j\neq i}^{N-1} \frac{T - x_j}{x_i - x_j}$$

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Through careful choice of $\{x_j\}_{j\in[N]}$, we can reduce the complexity of interpolation, multiplication, or other complex operations to $O(N \log N)$. **Spoiler:** we will use the Nth roots of unity domain $\Omega = \{\omega^j\}_{j\in[N]}$. Let us see why it helps.

Roots of Unity

Multiplicative Subgroup

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Number of Elements

The number of elements in $\mathbb{F}_p^\times$ is $p - 1$.

Primitive Root

Theorem

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$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1.$$

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The set $\mathbb{F}_p^\times$ is not useful on its own. However, we can consider the following set, called r**-th roots of unity**:

$$\Omega_r = \{\omega \in \mathbb{F}_p^\times : \omega^r = 1\} \subset \mathbb{F}_p^\times.$$

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Question. When does such a cyclic group exist?

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Yet another note

Typically, we would need r to be the power of two. We will see why in the NTT section.

Complex Analysis Interpretation

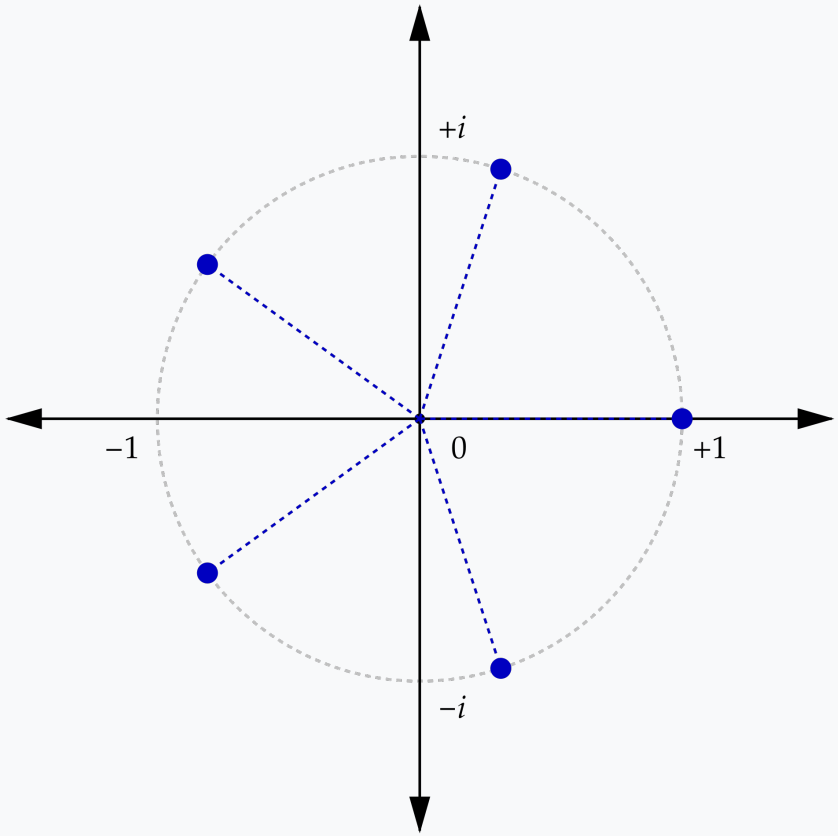


Figure: Visualization of the roots of unity $\Omega_5 = \{z \in \mathbb{C} : z^5 = 1\}$.

On the complex plane, the generator of the r-th roots of unity Ω_r is given by $\zeta_r = e^{2\pi i/r}$. In a finite field, we do not have such a luxury.

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Vanishing Polynomial over $\mathbb{R}$



Figure: Vanishing polynomial $p(x) = (x-1)(x-2)(x-4)$ of $D = \{1, 2, 4\}$

Barycentric Interpolation

Now, let us come back to the interpolation problem $p(x_j) = a_j$ for $j \in [N]$. Introduce $\gamma(T) = \prod_{j=0}^{N-1}(T - x_j)$.

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The Lagrange basis polynomial ℓ_j can be rewritten as:

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Takeaway: We can interpolate in $O(N)$ operations.

Number Theoretic Transform

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Definition (NTT)

Suppose $p(T) = \sum_{j=0}^{N-1} p_j T^j$. The **Number Theoretic Transform** (NTT) of p is defined as evaluations of p at the N-th roots of unity:

$$\mathrm{NTT}(\mathbf{p}) = \left(p(\omega^0), p(\omega^1), \ldots, p(\omega^{N-1})\right).$$

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Why? Well... $m(\omega^j) = p(\omega^j)q(\omega^j)$:/

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In its essence, we solve the interpolation problem:

$$m(\omega^j) = \hat{m}_j, \quad j \in [N], \quad \textbf{Goal}\text{: find coefficients } m_0, \ldots, m_{N-1}$$

Punchline

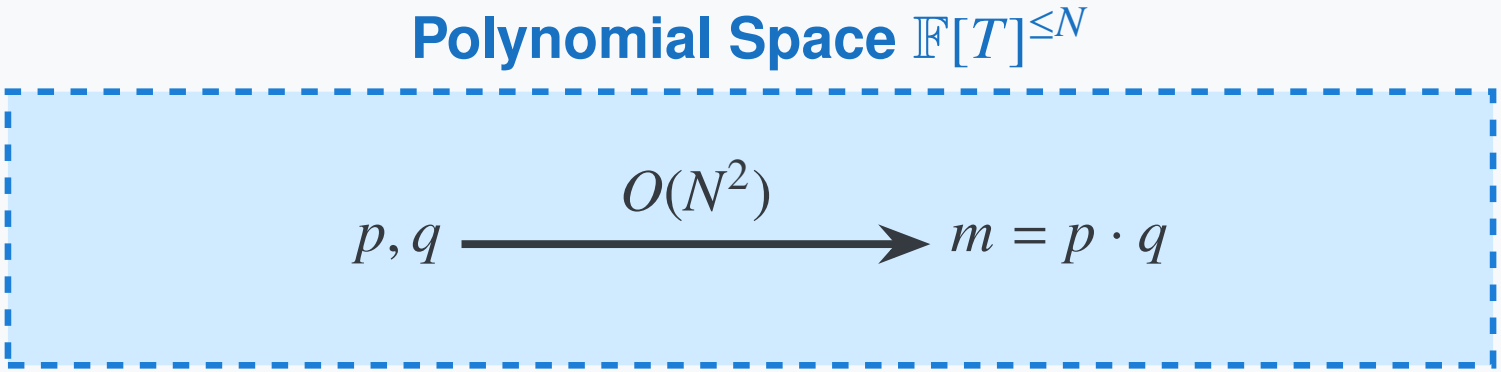


Figure: Illustration of the NTT Algorithm

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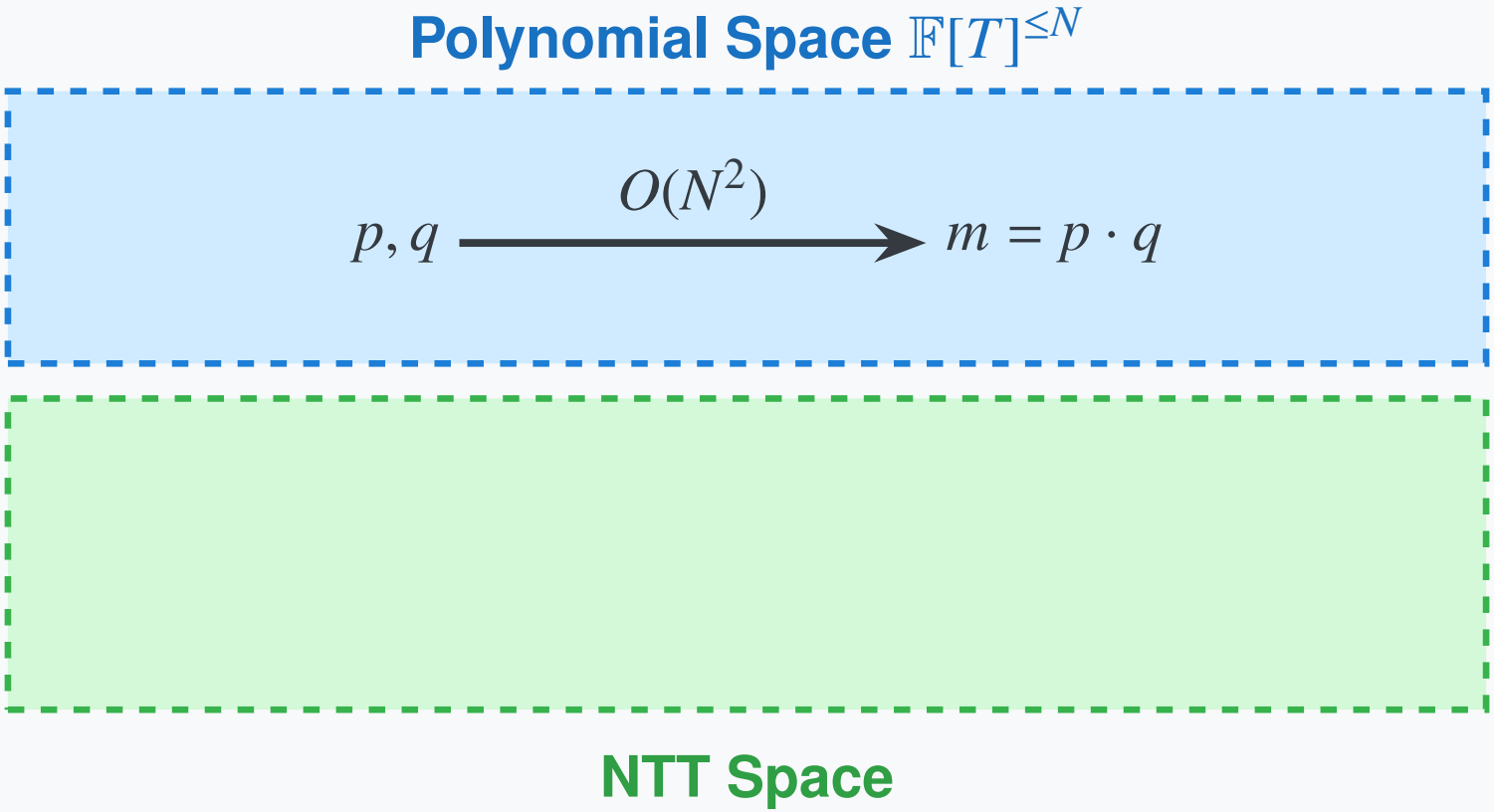


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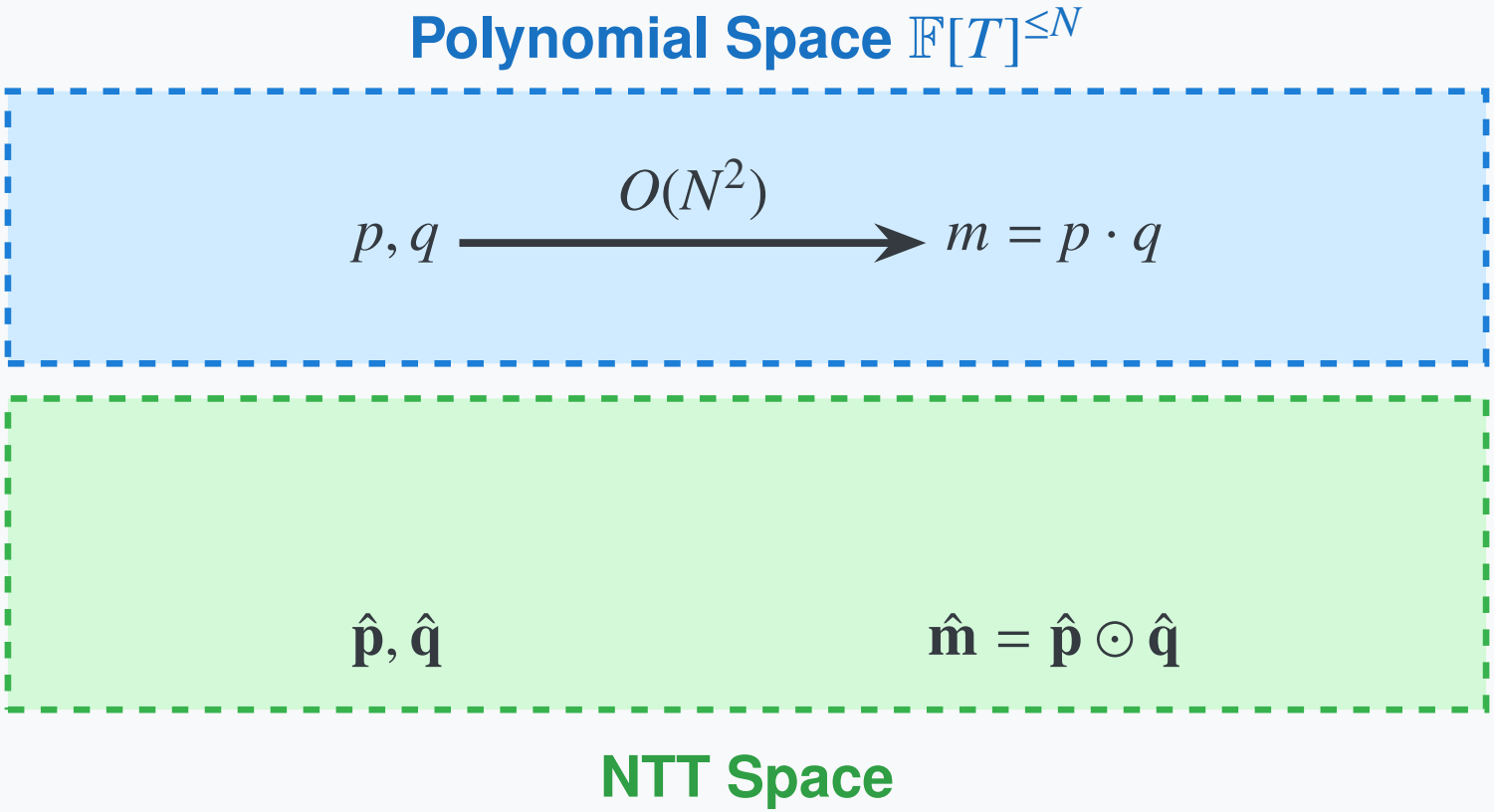


Figure: Illustration of the NTT Algorithm

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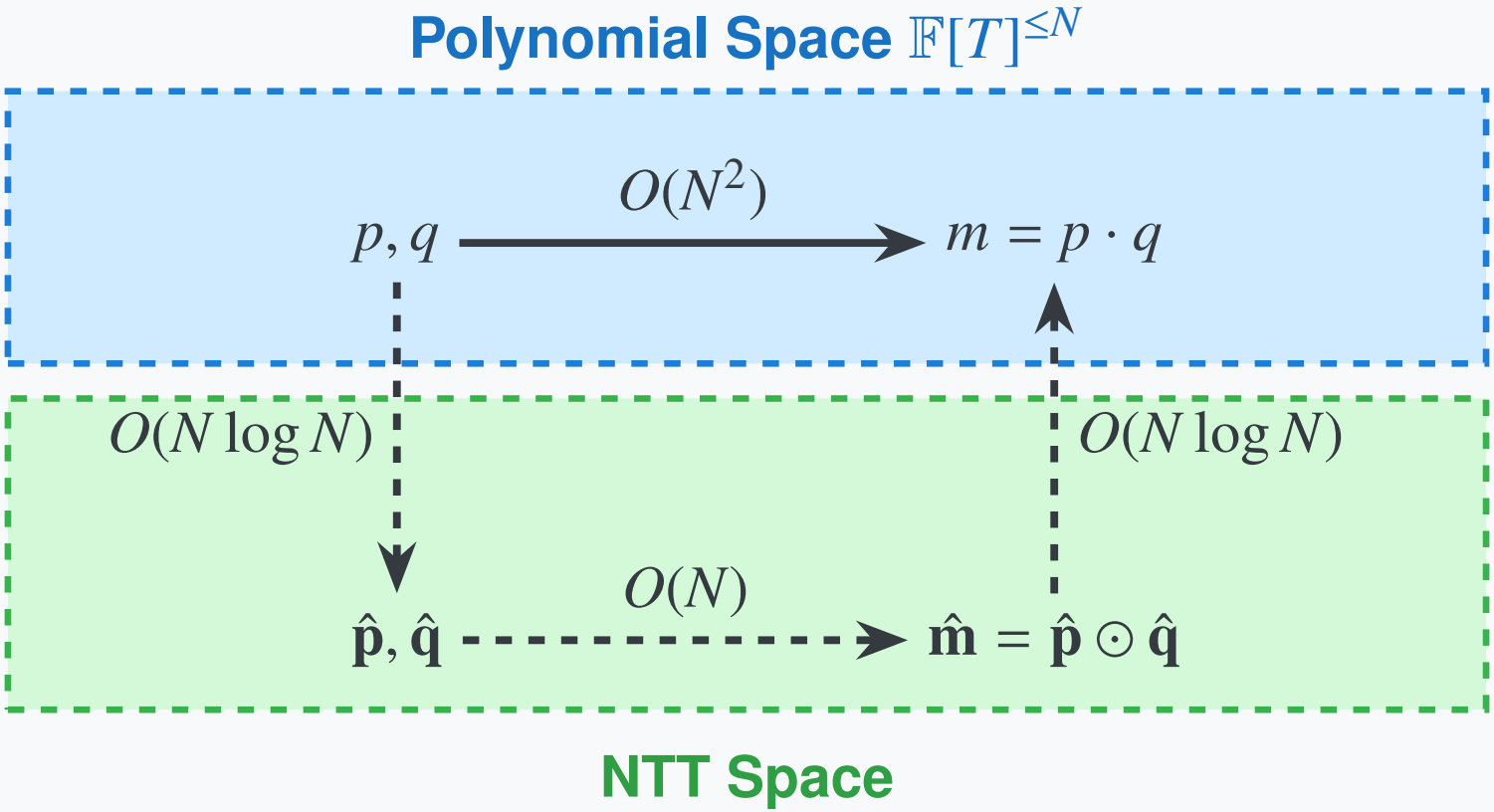


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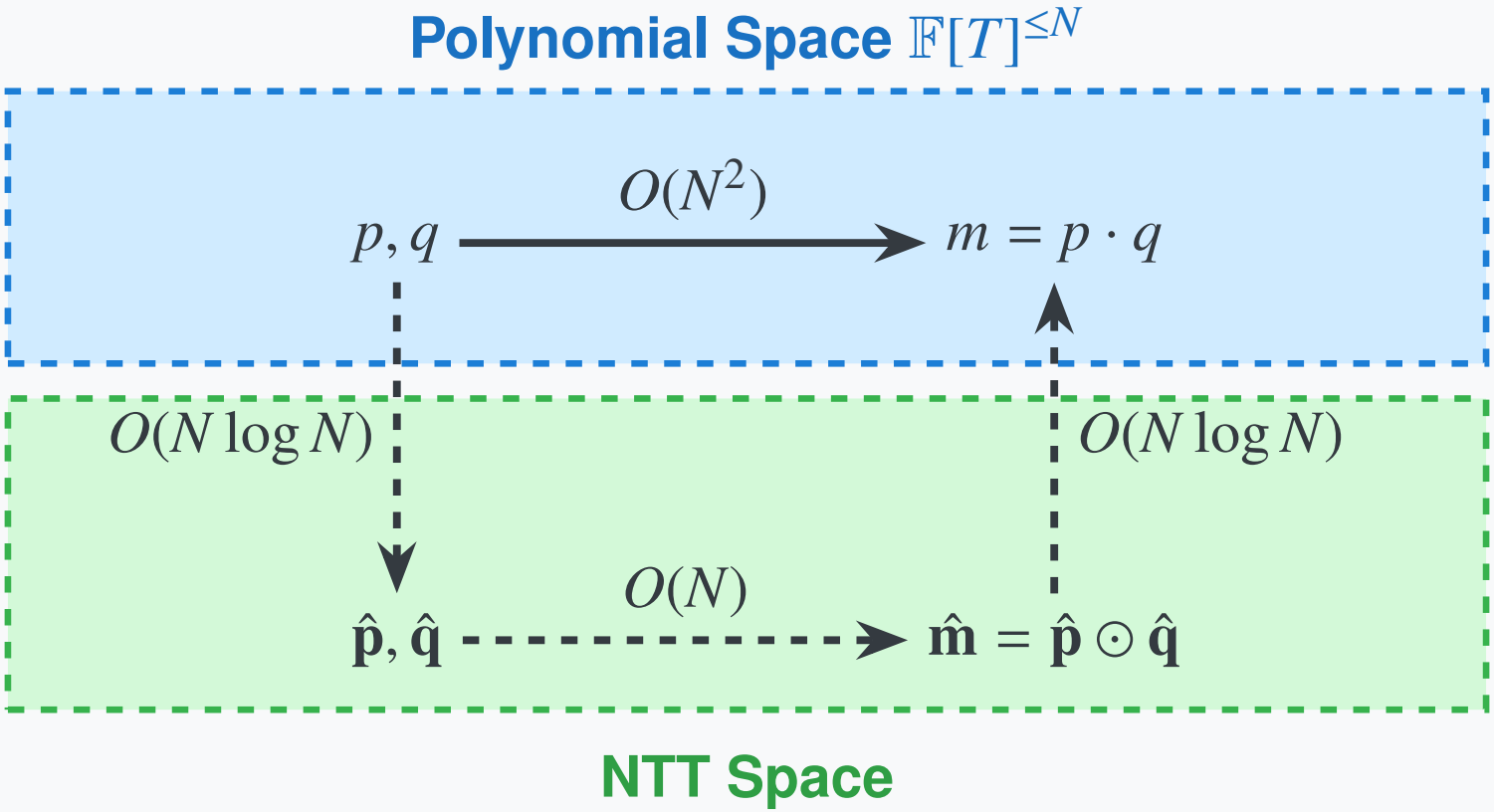


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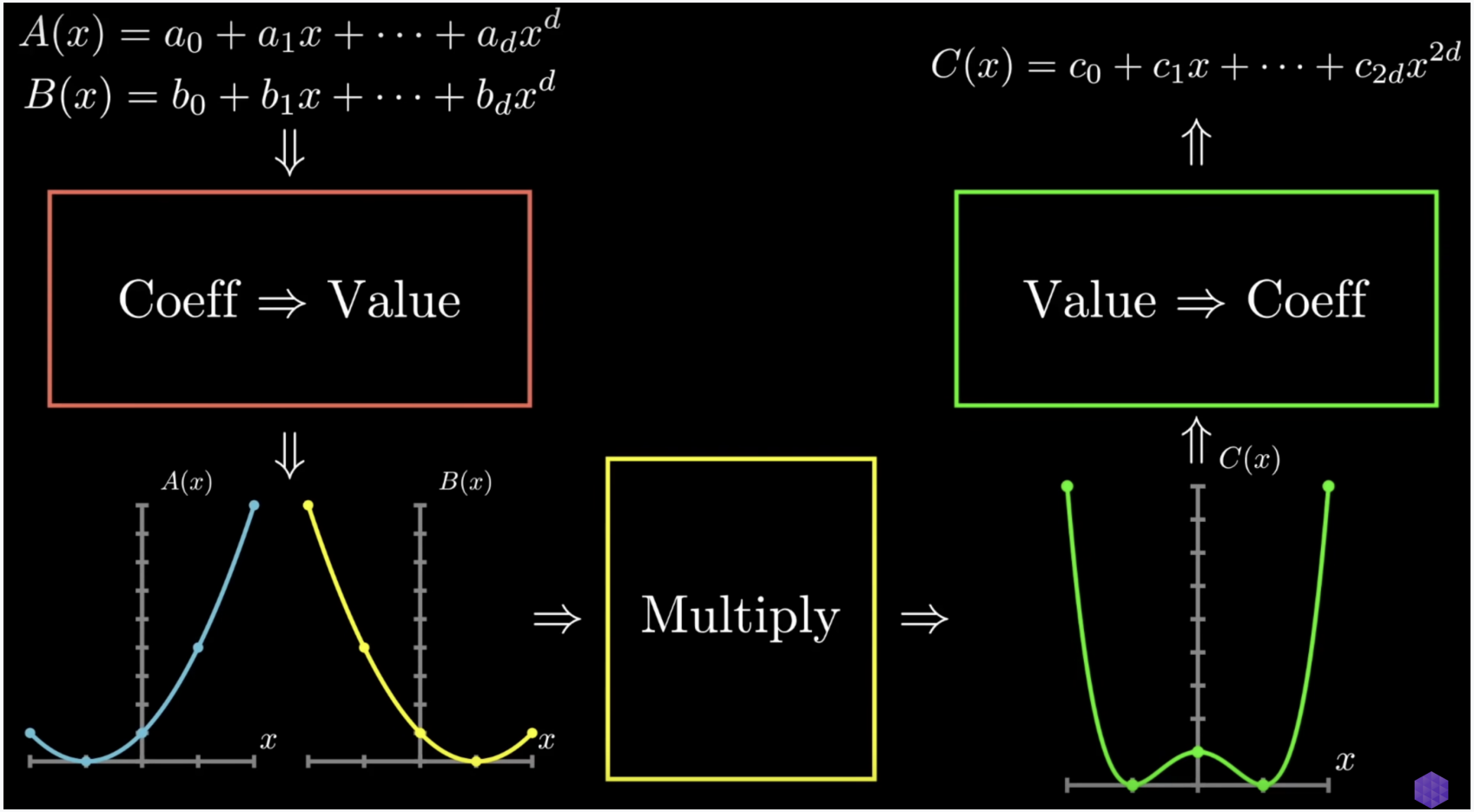


Figure: Illustration of the FFT Algorithm. Taken from “The Fast Fourier Transform (FFT): Most Ingenious Algorithm Ever?”

Details

When NTT works?

Note

For NTT to work, we will impose the following requirements on our setup:

1. The field $\mathbb{F}_p$ should have 2^k-roots of unity for sufficiently many k. In other words, $p = p' \cdot 2^m + 1$ with *small* $p' \in \mathbb{N}$.

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- **BabyBear prime** $p = 15 \cdot 2^{27} + 1$ is NTT-friendly: the order of multiplicative group is $15 \cdot 2^{27}$, so $2^k \mid 15 \cdot 2^{27}$ for all $k \leq 27$.

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- **Mersenne prime** $p = 2^{31} - 1$ is not NTT-friendly: the order of multiplicative group is $2^{31} - 2 = 2 \times (2^{30} - 1)$.

Why NTT takes quasilinear complexity?

Recall that we need to evaluate N expressions:

$$p(\omega^j) = \sum_{i=0}^{N-1} p_i (\omega^j)^i = \sum_{i=0}^{N-1} p_i \omega^{ij}, \quad j \in [N].$$

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$$p(\omega^j) = \sum_{i=0}^{2^r-1} p_i \omega^{ij} = \sum_{i=0}^{2^{r-1}-1} p_{2i} \omega^{2ij} + \sum_{i=0}^{2^{r-1}-1} p_{2i+1} \omega^{j(2i+1)}$$

$$= \sum_{i=0}^{2^{r-1}-1} p_{2i} (\omega^{2j})^i + \omega^j \sum_{i=0}^{2^{r-1}-1} p_{2i+1} (\omega^{2j})^i.$$

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Denote $p_E(T) = \sum_{i=0}^{2^{r-1}-1} p_{2i} T^i$ and $p_O(T) = \sum_{i=0}^{2^{r-1}-1} p_{2i+1} T^i$. Then,

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We need only $N/2$ evaluations from Ω of p_E and p_O. Note that:

$$p(\omega^{j+N/2}) = p_E(\omega^{2j}) + \omega^j \omega^{N/2} p_O(\omega^{2j}).$$

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Fact #2

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- We need to evaluate them at $N/2$ points.

Thus, we shrink the problem size by half at each step.

Algorithm Summarized

Algorithm 1: Number Theoretic Transform (NTT)

Input : Polynomial $p(T) = \sum_{j=0}^{N-1} p_j T^j$

Output : Vector of evaluations $\mathrm{NTT}(\mathbf{p}, \omega)$ at $\Omega = \{\omega^j\}_{j \in [N]}$

1 **if** $N = 1$ **then**
  Return : (p_0)
2 **end**
3 $H \leftarrow N/2$ /* Compute the domain half-size */
4 $\mathbf{p}_E \leftarrow (p_0, p_2, \ldots, p_{N-2})$ /* Find even-indexed coefficients */
5 $\mathbf{p}_O \leftarrow (p_1, p_3, \ldots, p_{N-1})$ /* Find odd-indexed coefficients */
6 $\mathbf{y}_E \leftarrow \mathrm{NTT}(\mathbf{p}_E, \omega^2)$ /* Compute NTT for even polynomial via $\frac{N}{2}$th primitive root ω^2 */
7 $\mathbf{y}_O \leftarrow \mathrm{NTT}(\mathbf{p}_O, \omega^2)$ /* Compute NTT for odd polynomial via $\frac{N}{2}$th primitive root ω^2 */

Return : $(y_0, \ldots, y_{N-1})$ with $y_j = y_{E,\, j \bmod H} + \omega^j y_{O,\, j \bmod H}$

Inverse NTT

Theorem

The Inverse NTT can be computed in the same way as NTT, but with the inverse primitive root ω^{-1}:

$$p_j = \frac{1}{N} \sum_{i=0}^{N-1} \omega^{-ij} \hat{p}_i$$

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Conclusion

To compute $m(T) = p(T)q(T)$, simply use the following:

$$\mathbf{m} = \mathsf{INTT}(\mathsf{NTT}(\mathbf{p}) \odot \mathsf{NTT}(\mathbf{q}))$$

The total complexity remains $O(N \log N)$.

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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